

# Two Apéry-transform supercongruences in OEIS A267220

A coefficient proof, a parameter refinement,  
and the dyadic correction

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## Abstract

Two formulas contributed by Peter Bala to OEIS A267220 on October 17, 2024 assert second-order supercongruences for diagonal powers of the exponential transform of the classical Apéry numbers and of its associated reversion transform. Both formulas are true for every integer value of the parameter. We prove them by establishing the second-order Gauss congruences for the Apéry numbers directly from their binomial sum, then giving a coefficient-level proof of preservation under normalized framing. For odd primes the result gains an additional factor of the prime dividing the parameter, and the restriction to primes at least five can be removed. The second family satisfies the original unsigned congruence also at the prime two; the first family requires an explicit sign correction at the first dyadic descent. We prove the full signed statement, treat zero and negative parameters, and give exact counterexamples to two tempting stronger assertions. The general framing-integrality mechanism is established prior work of Schwarz, Vologodsky, and Walcher, not a novelty claim of this note. The purpose here is to resolve the specific OEIS assertions with a complete, directly checkable argument. We also give a second, independent proof of the framing step, as one exact Frobenius-defect identity in the framed coordinate, and we exhibit an infinite family of binomial-moment seeds for which the entire argument goes through unchanged. Two standard-library Python programs and their exact verification records accompany the article.

**Scope and status.** This article supplies mathematical proofs, not an inference from numerical experiments. The general framing principle is prior work, not a novelty claim. The specific OEIS applications and refinements are proved explicitly here. This is an unrefereed research exposition; no proof-assistant verification or OEIS submission is claimed.

## 1 The two conjectures and the result

Let

$$a_n = \sum_{k=0}^n \binom{n}{k}^2 \binom{n+k}{k}^2 \quad (n \geq 0) \quad (1.1)$$

be the classical Apéry numbers, OEIS A005259 [2]. Their initial values are

$$1, 5, 73, 1445, 33001, 819005, 21460825, \dots$$

The two series relevant to the problem are

$$A(x) = \exp\left(\sum_{n \geq 1} \frac{a_n}{n} x^n\right), \quad F(x) = \frac{x}{\text{Rev}(xA(x))}. \quad (1.2)$$

Here  $\text{Rev}$  denotes compositional inversion at zero. Since  $xA(x) = x + O(x^2)$ , this inverse exists uniquely over  $\mathbb{Q}[[x]]$ . The quotient defining  $F$  is a well-defined formal power series with constant coefficient one. Integrality of both series will be proved below; it is not assumed in the argument.

The coefficients of  $A$  form OEIS A267220. The entry includes two conjectures, both explicitly attributed to Peter Bala and dated October 17, 2024 [1]. In our notation they concern

$$u_m(n) = [x^n]A(x)^{mn}, \quad v_m(n) = [x^n]F(x)^{mn} \quad (m \in \mathbb{Z}, n \geq 0), \quad (1.3)$$

and assert

$$\begin{aligned} u_m(np^r) &\equiv u_m(np^{r-1}) \pmod{p^{2r}}, \\ v_m(np^r) &\equiv v_m(np^{r-1}) \pmod{p^{2r}} \end{aligned} \quad (1.4)$$

for every prime  $p \geq 5$  and all positive integers  $n, r$ . These assertions were still labeled conjectural when the entry was checked on September 20, 2026. An older, different integrality comment in that entry already cites a proof; it is not being presented here as an open problem. That older comment, with its 2020 attribution to Beukers, is not one of the two targeted conjectures.

The notation  $a_n$  in this article always denotes the Apéry numbers, *not* the coefficients of A267220. In particular, these are two related but different sequences. Also,

$$u_m(0) = v_m(0) = 1$$

for every  $m$ , whereas  $u_0(n) = v_0(n) = 0$  for  $n \geq 1$ . This distinction matters when normalizing by the parameter.

**Theorem 1.1** (Resolution and refinements). *Both congruences (1.4) hold for every integer  $m$ . More precisely, let  $N \geq 1$ , let  $p \mid N$  be prime, and put  $R = \nu_p(N) \geq 1$ .*

(a) *If  $p$  is odd and  $m \neq 0$ , then*

$$\begin{aligned} \nu_p(u_m(N) - u_m(N/p)) &\geq 2R + \nu_p(m), \\ \nu_p(v_m(N) - v_m(N/p)) &\geq 2R + \nu_p(m). \end{aligned} \quad (1.5)$$

(b) *The unsigned congruence for the second family holds at every prime, including 2:*

$$v_m(N) \equiv v_m(N/p) \pmod{p^{2R}}. \quad (1.6)$$

(c) *For every prime  $p$  and  $m \neq 0$ , the signed refinements are*

$$\begin{aligned} (-1)^{mN} u_m(N) &\equiv (-1)^{mN/p} u_m(N/p) \pmod{p^{2R+\nu_p(m)}}, \\ (-1)^{(m-1)N} v_m(N) &\equiv (-1)^{(m-1)N/p} v_m(N/p) \pmod{p^{2R+\nu_p(m)}}. \end{aligned} \quad (1.7)$$

*For  $m = 0$ , all the positive-index terms in these statements vanish.*

We use  $\nu_p(0) = +\infty$ , and  $\nu_p(m) = \nu_p(|m|)$  for nonzero negative integers. Substituting  $N = np^r$  gives  $R = r + \nu_p(n)$ , so part (a) proves a modulus at least as strong as the one in OEIS even when  $p$  divides  $n$ . The theorem does not tacitly assume  $p \nmid n$  or  $p \nmid m$ .

## What is established here, and what is prior work?

The nonlinear preservation principle behind the proof is the *integrality of framing* for 2-functions. It is proved in general by Schwarz, Vologodsky, and Walcher [3, 4]; their 2017 paper includes a local coefficient proof. Our proof uses that known mechanism in an explicit normalization adapted to the two OEIS families. No claim is made that the general preservation theorem, or its connection to dilogarithms, is new. Nor does a conjectural label in OEIS establish that no earlier specialist has deduced this particular application. Nor does an entry's conjectural label establish that no proof exists elsewhere.

The contribution of this note is a self-contained resolution of the stated formulas, with all parameter values and the prime two handled explicitly, together with reproducible computations. The proof does not appeal to a computer search as a substitute for an all-index argument. Its dependency chain is

$$\begin{aligned} \text{Apéry binomial sum} &\implies \text{second-order Gauss congruences} \implies \text{integral } A, \\ \text{normalized coefficient transform} &\implies \text{framing congruences} \implies \text{both OEIS assertions.} \end{aligned}$$

## 2 Formal arithmetic and the dilogarithm criterion

Everything below takes place in formal power series rings. No analytic convergence or choice of a complex logarithm is involved. Set

$$\delta = x \frac{d}{dx}, \quad \text{CT} \left( \sum_{j \geq j_0} q_j x^j \right) = q_0.$$

The constant-term functional is applied only to Laurent series bounded below. It satisfies

$$\text{CT}(\delta G) = 0, \quad \text{CT}(G \delta J) = -\text{CT}(J \delta G). \quad (2.1)$$

Each relevant constant term depends on only finitely many coefficients.

For a prime  $p$ , write

$$\mathbb{Z}_{(p)} = \{a/b \in \mathbb{Q} : p \nmid b\}.$$

A congruence between elements of  $\mathbb{Z}_{(p)}$  has its usual valuation meaning; a congruence between series means the congruence of every coefficient. Working in this localization is sufficient: completeness is unnecessary.

**Definition 2.1.** An integer sequence  $(\alpha_n)_{n \geq 1}$  satisfies the *second-order Gauss congruences* if

$$\alpha_N \equiv \alpha_{N/p} \pmod{p^{2\nu_p(N)}} \quad (2.2)$$

for every prime  $p$  and every positive  $N$  divisible by  $p$ .

**Lemma 2.2** (Möbius and dilogarithm criterion). *For an integer sequence  $(\alpha_n)_{n \geq 1}$ , the following are equivalent:*

- (i) *The sequence satisfies (2.2).*
- (ii) *The numbers*

$$c_n = \frac{1}{n^2} \sum_{d|n} \mu(n/d) \alpha_d \quad (2.3)$$

*are integers for every  $n \geq 1$ .*

(iii) *There is an expansion with integer coefficients*

$$\sum_{n \geq 1} \frac{\alpha_n}{n^2} x^n = \sum_{d \geq 1} c_d \operatorname{Li}_2(x^d), \quad \operatorname{Li}_2(z) = \sum_{j \geq 1} \frac{z^j}{j^2}. \quad (2.4)$$

The  $c_d$  in (iii) are uniquely given by (2.3).

*Proof.* Fix  $n = p^r h$  with  $p \nmid h$ . In the numerator of (2.3), the only nonzero terms have divisor  $p$ -adic exponent  $r$  or  $r - 1$ . Pairing them gives

$$\sum_{d|n} \mu(n/d) \alpha_d = \sum_{e|h} \mu(h/e) (\alpha_{p^r e} - \alpha_{p^{r-1} e}).$$

Under (i), this numerator is divisible by  $p^{2r}$ . Doing this for every prime divisor of  $n$  proves divisibility by  $n^2$ , hence (ii).

Möbius inversion gives

$$\alpha_n = \sum_{d|n} d^2 c_d. \quad (2.5)$$

If every  $c_d$  is an integer, subtracting the formula for  $\alpha_{N/p}$  from that for  $\alpha_N$  leaves only divisors  $d$  with  $\nu_p(d) = \nu_p(N)$ . Every surviving term contains  $p^{2\nu_p(N)}$ , proving (i). Finally, coefficient extraction from the right side of (2.4) gives  $n^{-2} \sum_{d|n} d^2 c_d$ , proving the equivalence with (iii).  $\square$

One immediate benefit of this stronger integrality criterion is an Euler product. Define

$$W(x) = \sum_{n \geq 1} \frac{\alpha_n}{n^2} x^n, \quad H(x) = \delta W(x) = \sum_{n \geq 1} \frac{\alpha_n}{n} x^n.$$

Taking  $\delta$  of (2.4) and then exponentiating gives

$$\exp H(x) = \prod_{d \geq 1} (1 - x^d)^{-dc_d} \in 1 + x\mathbb{Z}[[x]]. \quad (2.6)$$

This identity is formal: to compute the coefficient of  $x^n$ , only factors with  $d \leq n$  are needed. The exponents may be negative or positive; any integer power of a unit in  $1 + x\mathbb{Z}[[x]]$  is integral.

A second formulation will be central to the coefficient proof.

**Lemma 2.3** (Integral Frobenius defect). *If  $(\alpha_n)$  satisfies the second-order Gauss congruences, then for every prime  $p$  the series*

$$E_p(x) = W(x) - \frac{1}{p^2} W(x^p) \quad (2.7)$$

*belongs to  $x\mathbb{Z}_{(p)}[[x]]$ .*

*Proof.* If  $p \nmid n$ , the coefficient is  $\alpha_n/n^2 \in \mathbb{Z}_{(p)}$ . If  $p \mid n$ , it is  $(\alpha_n - \alpha_{n/p})/n^2$ , again in  $\mathbb{Z}_{(p)}$  by the hypothesis.  $\square$

Although  $W$  and  $H = \delta W$  may have denominators, all *higher* derivatives are integral with no hypothesis at all:

$$\delta^k W(x) = \sum_{n \geq 1} n^{k-2} \alpha_n x^n \in x\mathbb{Z}[[x]] \quad (k \geq 2). \quad (2.8)$$

The distinction between the first derivative and the higher derivatives is what makes the cancellation in Section 8.3 work.

### 3 The required Apéry congruence from the binomial sum

We now return to the sequence  $a_n$  in (1.1). Only a second-order congruence is needed. It admits a short proof that works uniformly at all primes, including 2 and 3. It is classical that the Apéry numbers satisfy the stronger order-three congruences at primes at least 5; the historical results of Gessel and Coster are reviewed in [6], which also records the standard recurrence. That material is background and context only. Nothing below depends on it: the required order-two input is proved here directly from the binomial sum.

**Proposition 3.1.** *For every prime  $p$  and positive integer  $N$  divisible by  $p$ ,*

$$a_N \equiv a_{N/p} \pmod{p^{2\nu_p(N)}}. \quad (3.1)$$

*Proof.* Put  $R = \nu_p(N)$  and

$$P(N, k) = \binom{N}{k} \binom{N+k}{k}.$$

When  $p \nmid k$ , the identity

$$\binom{N}{k} = \frac{N}{k} \binom{N-1}{k-1}$$

shows that  $\nu_p(P(N, k)) \geq R$ . Such a summand  $P(N, k)^2$  therefore vanishes modulo  $p^{2R}$ .

It remains to compare the summands with  $k = p\ell$ . For  $k \geq 1$ , an exact product identity is

$$P(N, k) = (-1)^{k-1} \frac{N}{k} \left(1 + \frac{N}{k}\right) \prod_{j=1}^{k-1} \left(1 - \frac{N^2}{j^2}\right). \quad (3.2)$$

To see this, write the binomial product as

$$\frac{N(N+k)}{k^2} \prod_{j=1}^{k-1} \frac{(N-j)(N+j)}{j^2}$$

and extract the signs. Separating the factors with  $p \mid j$  in (3.2) yields

$$P(N, p\ell) = (-1)^{(p-1)\ell} P(N/p, \ell) \prod_{\substack{1 \leq j < p\ell \\ p \nmid j}} \left(1 - \frac{N^2}{j^2}\right). \quad (3.3)$$

Each factor in the remaining product belongs to  $1 + p^{2R}\mathbb{Z}_{(p)}$ . Squaring removes the sign even when  $p = 2$ , so

$$P(N, p\ell)^2 \equiv P(N/p, \ell)^2 \pmod{p^{2R}}.$$

The  $k = 0$  summand is one on both sides. Summing over  $0 \leq k \leq N$  now gives exactly (3.1).  $\square$

Consequently, Lemma 2.2 applies to  $a_n$ , and the series  $A$  in (1.2) is integral by (2.6). Its primitive dilogarithm coefficients begin

$$\begin{aligned} c_1 &= 5, & c_2 &= 17, & c_3 &= 160, \\ c_4 &= 2058, & c_5 &= 32760, & c_6 &= 596092. \end{aligned} \quad (3.4)$$

For example,  $c_4 = (a_4 - a_2)/16 = 2058$  and  $c_6 = (a_6 - a_3 - a_2 + a_1)/36 = 596092$ . Thus

$$A(x) = (1-x)^{-5}(1-x^2)^{-34}(1-x^3)^{-480}(1-x^4)^{-8232} \dots$$

The arithmetic input to the rest of the proof is now complete. We have not used a stronger Apéry supercongruence as a black box.

## 4 An integral normalization of the two transforms

From now on set

$$W = \sum_{n \geq 1} \frac{a_n}{n^2} x^n, \quad H = \delta W = \log A, \quad D = \delta H = \sum_{n \geq 1} a_n x^n. \quad (4.1)$$

The series  $D$  has constant coefficient zero; in particular, it is the Apéry generating series with its initial term  $a_0 = 1$  removed.

For every  $m \in \mathbb{Z}$ , define  $X_m(y)$  by

$$X_m(y) = yA(X_m(y))^m, \quad \text{equivalently} \quad y = xA(x)^{-m}. \quad (4.2)$$

The inverse exists uniquely in  $y + y^2\mathbb{Z}[[y]]$ : a series  $x + O(x^2)$  with integer coefficients has an integer compositional inverse, as is seen by solving successively for its coefficients. This includes negative  $m$ , since  $A$  and every integer power of  $A$  are integral units.

Define the normalized transform by

$$\mathcal{B}_m(y) = \frac{D(X_m(y))}{1 - mD(X_m(y))} = \sum_{n \geq 1} b_m(n) y^n. \quad (4.3)$$

The denominator is a unit of constant coefficient one. Therefore

$$b_m(n) \in \mathbb{Z} \quad (m \in \mathbb{Z}, n \geq 1), \quad b_0(n) = a_n. \quad (4.4)$$

This integral definition is preferable to starting with a quotient by  $m$ . In particular, it remains valid when a prime under consideration divides  $m$ .

**Proposition 4.1** (Coefficient normalization). *For  $m \neq 0$  and  $n \geq 1$ ,*

$$b_m(n) = \frac{1}{m} [x^n] A(x)^{mn}. \quad (4.5)$$

*Consequently,  $u_m(n) = mb_m(n)$  for every integer  $m$  and every  $n \geq 1$ .*

*Proof.* Logarithmic differentiation of  $y = xA(x)^{-m}$  gives

$$\frac{d \log y}{d \log x} = 1 - mD(x),$$

so the chain rule implies

$$\delta_y H(X_m(y)) = \mathcal{B}_m(y). \quad (4.6)$$

The formal Lagrange inversion formula (proved in Appendix A) gives

$$[y^n] H(X_m(y)) = \frac{1}{n} [x^{n-1}] H'(x) A(x)^{mn}.$$

For  $m \neq 0$ , use  $(A^{mn})' = mnH'A^{mn}$ . The last display becomes

$$[y^n] H(X_m(y)) = \frac{1}{mn} [x^n] A(x)^{mn}.$$

Multiplication by  $n$ , as in (4.6), proves the assertion. At  $m = 0$ , the identity  $u_0(n) = 0 = mb_0(n)$  holds separately.  $\square$

The relation between the two conjectures is particularly simple after this normalization.

**Proposition 4.2** (The reversion family is a shifted transform). *The series  $F$  is integral and satisfies*

$$F(y) = A(X_{-1}(y)). \quad (4.7)$$

For every integer  $m$  and every  $n \geq 1$ ,

$$v_m(n) = mb_{m-1}(n). \quad (4.8)$$

In particular,  $v_1(n) = a_n$ .

*Proof.* Equation (4.2) with  $m = -1$  says  $X_{-1}(y)A(X_{-1}(y)) = y$ , hence  $X_{-1} = \text{Rev}(xA(x))$ . Dividing by  $X_{-1}$  proves (4.7), including its integrality.

Apply Lagrange inversion to  $A(X_{-1}(y))^{mn}$ . For  $n \geq 1$  it gives

$$[y^n]A(X_{-1}(y))^{mn} = m[x^{n-1}]A'(x)A(x)^{(m-1)n-1}. \quad (4.9)$$

If  $m \neq 1$ , differentiating  $A^{(m-1)n}$  turns this into

$$\frac{m}{m-1}[x^n]A(x)^{(m-1)n} = mb_{m-1}(n).$$

For  $m = 1$ , the right side of (4.9) is  $[x^{n-1}]A'/A = [x^{n-1}]H' = a_n = b_0(n)$ . Thus the apparent singularity at  $m = 1$  is removable, and the asserted formula holds for all  $m$ .  $\square$

**Remark 4.3** (Why division modulo a prime is not enough). The formula  $v_m(n) = \frac{m}{m-1}u_{m-1}(n)$  for  $m \neq 1$  is an identity over  $\mathbb{Q}$ , but it cannot safely be used to transfer a congruence by dividing by  $m-1$  when  $p \mid m-1$ . Similarly, dividing a congruence for  $u_m$  by  $m$  may lose powers of  $p$ . The proof below establishes the congruence for the integer  $b_m$  itself. This is why no exceptional parameter classes need to be excluded.

## 5 The normalized framing congruence at odd primes

The next theorem is the coefficient form of the established framing-integrality principle. Its proof is included rather than invoked from the general theory [3, 4].

**Theorem 5.1** (Normalized framing congruences). *Let  $b_m$  be as in (4.3). For every integer  $m$ , define*

$$\beta_m(n) = (-1)^{mn}b_m(n) \quad (n \geq 1). \quad (5.1)$$

*Then  $\beta_m$  satisfies the second-order Gauss congruences at every prime. Equivalently, for  $p \mid N$  and  $R = \nu_p(N)$ ,*

$$(-1)^{mN}b_m(N) \equiv (-1)^{mN/p}b_m(N/p) \pmod{p^{2R}}. \quad (5.2)$$

*For odd  $p$  this is the ordinary unsigned congruence for  $b_m$ . At  $p = 2$  the unsigned congruence also holds whenever  $m$  is even or  $R \geq 2$ .*

We prove the odd-prime assertion in this section and the remaining cases in Section 6. These arguments use only the integer coefficients  $a_n$  and their second-order Gauss congruences. Thus the same proof establishes the theorem for any input sequence satisfying Definition 2.1.

### 5.1 An exact constant-term expression

Fix  $p \mid N$ , put  $R = \nu_p(N)$ , and first take  $m \neq 0$ . For brevity write  $E = E_p$ . Lemma 2.3 gives

$$E = W(x) - p^{-2}W(x^p) \in x\mathbb{Z}_{(p)}[[x]].$$

Differentiating and using  $H = \delta W$  yields

$$\delta E = H(x) - p^{-1}H(x^p), \quad A(x^p) = A(x)^p \exp(-p\delta E). \quad (5.3)$$

Introduce the integral Laurent series

$$K(x) = x^{-N}A(x)^{mN} \in x^{-N}\mathbb{Z}[[x]]. \quad (5.4)$$

Since substitution  $x \mapsto x^p$  turns the coefficient of  $x^{N/p}$  into the coefficient of  $x^N$ , Proposition 4.1 and (5.3) give the exact identity

$$b_m(N) - b_m(N/p) = \frac{1}{m} \text{CT}(K(1 - \exp(-mN\delta E))). \quad (5.5)$$

Expanding the exponential gives

$$b_m(N) - b_m(N/p) = \sum_{k \geq 1} \frac{(-1)^{k+1} m^{k-1} N^k}{k!} \text{CT}(K(\delta E)^k). \quad (5.6)$$

Only  $k \leq N$  can contribute, because  $\delta E$  has no constant term. Thus no question about convergence of the infinite displayed sum arises. The factor  $m$  has disappeared from the denominator of each summand: this is the crucial normalization when  $p \mid m$ .

### 5.2 The linear term gains a second power of the index

The  $k = 1$  term seems initially to contain only  $N$ , not  $N^2$ . Constant-term integration by parts supplies the missing factor. Namely,

$$\delta K = N(mD - 1)K, \quad (5.7)$$

and therefore

$$\begin{aligned} N \text{CT}(K\delta E) &= -N \text{CT}(E\delta K) \\ &= N^2 \text{CT}(K(1 - mD)E) \in p^{2R}\mathbb{Z}_{(p)}. \end{aligned} \quad (5.8)$$

All coefficients inside the final constant term are  $p$ -integral. This identity, rather than a coarse exponential estimate alone, is what produces a second-order congruence.

### 5.3 The nonlinear terms at odd primes

For an odd prime and  $k \geq 2$ ,

$$\nu_p(k!) \leq k - 2. \quad (5.9)$$

For  $k = 2$  this is immediate. For  $k \geq 3$ , Legendre's formula gives  $\nu_p(k!) \leq (k-1)/(p-1) \leq k-2$ . Since  $\nu_p(m) \geq 0$  and  $R \geq 1$ ,

$$\begin{aligned} \nu_p\left(\frac{m^{k-1}N^k}{k!}\right) &\geq kR - (k-2) \\ &= 2R + (k-2)(R-1) \geq 2R. \end{aligned}$$

Also  $\text{CT}(K(\delta E)^k) \in \mathbb{Z}_{(p)}$ . Every nonlinear term in (5.6) is therefore divisible by  $p^{2R}$ . Together with (5.8), this proves

$$b_m(N) \equiv b_m(N/p) \pmod{p^{2R}} \quad (p \text{ odd, } m \neq 0). \quad (5.10)$$

For  $m = 0$ , this is precisely Proposition 3.1. Finally,  $N$  and  $N/p$  have the same parity when  $p$  is odd, so the two signs in (5.2) agree. This proves the odd-prime part of Theorem 5.1.

## 6 The prime two and the exact sign correction

The odd-prime factorial bound fails at  $p = 2$ . Replacing it indiscriminately by the dyadic bound would lose information and would not prove the asserted modulus. There are three distinct cases. Throughout this section  $E = E_2$ ,  $K = x^{-N} A^{mN}$ , and  $R = \nu_2(N) \geq 1$ . The case  $m = 0$  is already covered by the input Apéry congruence.

### 6.1 Even parameters

Suppose  $m \neq 0$  is even. For  $k \geq 2$ , the bound  $\nu_2(k!) \leq k - 1$  gives

$$\nu_2 \left( \frac{m^{k-1} N^k}{k!} \right) \geq (k-1) + kR - (k-1) = kR \geq 2R.$$

The linear term is handled by (5.8). Hence

$$b_m(N) \equiv b_m(N/2) \pmod{2^{2R}} \quad (m \text{ even}). \quad (6.1)$$

The signs in (5.2) are both positive in this case.

### 6.2 Odd parameters beyond the first dyadic descent

Suppose  $m$  is odd and  $R \geq 2$ . For  $k \geq 3$ ,

$$\nu_2 \left( \frac{m^{k-1} N^k}{k!} \right) \geq kR - (k-1) \geq 2R.$$

The last inequality follows from  $(k-2)R \geq k-1$ , valid for  $R \geq 2$  and  $k \geq 3$ . The only problematic nonlinear term is  $k = 2$ , whose scalar coefficient  $mN^2/2$  has valuation  $2R-1$ .

Write  $E = \sum_{j \geq 1} e_j x^j$ , with  $e_j \in \mathbb{Z}_{(2)}$ . Reduction modulo two gives

$$(\delta E)^2 \equiv \sum_{\substack{j \geq 1 \\ j \text{ odd}}} e_j^2 x^{2j} \pmod{2}. \quad (6.2)$$

For odd  $j$ , applying CT  $\delta$  to  $Kx^{2j}$  and using (5.7) yields

$$2j \text{ CT}(Kx^{2j}) = N \text{ CT}(K(1 - mD)x^{2j}). \quad (6.3)$$

Since  $j$  is odd, the right side shows  $\text{CT}(Kx^{2j}) \in 2^{R-1}\mathbb{Z}_{(2)} \subseteq 2\mathbb{Z}_{(2)}$ . Equation (6.2) consequently implies

$$\text{CT}(K(\delta E)^2) \in 2\mathbb{Z}_{(2)}.$$

This supplies the missing factor two in the quadratic term. The linear term again follows from (5.8), proving

$$b_m(N) \equiv b_m(N/2) \pmod{2^{2R}} \quad (m \text{ odd, } R \geq 2). \quad (6.4)$$

Here  $N$  and  $N/2$  are both even, so this is also the signed congruence.

### 6.3 Odd parameters at the first dyadic descent

It remains to take  $m$  odd and  $N = 2h$  with  $h$  odd. The correct relation is now a *sum*:

$$b_m(2h) + b_m(h) \equiv 0 \pmod{4}. \quad (6.5)$$

Using the same derivation as in (5.5), but adding instead of subtracting, we obtain

$$b_m(N) + b_m(h) = \frac{1}{m} \text{CT}(K(1 + \exp(-mN\delta E))). \quad (6.6)$$

Let  $s_2(k)$  denote the number of ones in the binary expansion of  $k$ . Legendre's formula at two is

$$\nu_2(k!) = k - s_2(k).$$

As  $mN$  has valuation one, the coefficient  $(-mN)^k/k!$  has valuation  $s_2(k)$ . Modulo four, only positive exponents  $k$  that are powers of two survive. Each surviving scalar is congruent to two modulo four, since it has valuation exactly one. Thus, coefficientwise in  $\mathbb{Z}_{(2)}[[x]]$ ,

$$1 + \exp(-mN\delta E) \equiv 2 \left( 1 + \sum_{s \geq 0} (\delta E)^{2^s} \right) \pmod{4}. \quad (6.7)$$

The sum on the right is locally finite in degree.

We next identify its reduction modulo two. In  $\delta E$ , terms of even degree vanish modulo two. For odd  $j$ ,  $e_j = a_j/j^2$ , and hence

$$\delta E \equiv \sum_{j \text{ odd}} a_j x^j \pmod{2}.$$

Frobenius in characteristic two gives

$$\sum_{s \geq 0} (\delta E)^{2^s} \equiv \sum_{s \geq 0} \sum_{j \text{ odd}} a_j x^{j2^s} \pmod{2}.$$

The input congruences imply  $a_{j2^s} \equiv a_j \pmod{2}$ . Every positive integer is uniquely  $j2^s$  with  $j$  odd. Therefore

$$\sum_{s \geq 0} (\delta E)^{2^s} \equiv D(x) \pmod{2}. \quad (6.8)$$

Combining (6.6)–(6.8) gives

$$b_m(N) + b_m(h) \equiv \frac{2}{m} \text{CT}(K(1 + D)) \pmod{4}.$$

Because  $m$  is odd,  $1 + D \equiv 1 - mD \pmod{2}$ . But (5.7) and  $\text{CT} \delta K = 0$  give the exact identity

$$\text{CT}(K(1 - mD)) = 0. \quad (6.9)$$

It follows that  $\text{CT}(K(1 + D))$  is even, proving (6.5). Since  $(-1)^{mN} = 1$  and  $(-1)^{mh} = -1$ , this is exactly the missing case of (5.2). Theorem 5.1 is proved.

## 7 Deduction of both OEIS conjectures

We now prove Theorem 1.1 and make clear where the two families behave differently at two.

## 7.1 Odd primes and the parameter factor

For  $p$  odd, Theorem 5.1 and the identities

$$u_m(n) = mb_m(n), \quad v_m(n) = mb_{m-1}(n)$$

immediately imply

$$\begin{aligned} u_m(N) - u_m(N/p) &\in mp^{2R}\mathbb{Z}_{(p)}, \\ v_m(N) - v_m(N/p) &\in mp^{2R}\mathbb{Z}_{(p)}. \end{aligned}$$

This is (1.5). In particular, the originally stated range  $p \geq 5$  may be replaced by all odd primes.

Multiplying the all-prime signed congruence for  $b_m$  by  $m$  proves the first part of (1.7). Applying the same congruence to  $b_{m-1}$  and multiplying by  $m$  proves its second part. Notice that the sign parameter for  $v_m$  is  $m - 1$ , but the extra valuation comes from  $m$ , not from  $m - 1$ .

## 7.2 Why the second family works unsigned at two

When  $m$  is odd, the normalized parameter  $m - 1$  is even. Thus (6.1) immediately proves  $v_m(N) \equiv v_m(N/2) \pmod{2^{2R}}$ .

When  $m$  is even and  $R \geq 2$ , (6.4) gives the same result, in fact with the extra factor  $2^{\nu_2(m)}$ . It remains to consider a nonzero even  $m$  and  $N = 2h$  with  $h$  odd. Now  $m - 1$  is odd, so the signed theorem says

$$b_{m-1}(2h) + b_{m-1}(h) \in 4\mathbb{Z}.$$

Consequently,

$$\begin{aligned} v_m(2h) - v_m(h) &= m(b_{m-1}(2h) + b_{m-1}(h)) - 2mb_{m-1}(h) \\ &\in 2^{\nu_2(m)+1}\mathbb{Z} \subseteq 4\mathbb{Z}. \end{aligned} \tag{7.1}$$

This proves (1.6) in the last case. The zero parameter has already been separated, so Theorem 1.1 is complete.

## 7.3 A precise dyadic summary

For the first family, the unsigned parameter-refined bound

$$\nu_2(u_m(N) - u_m(N/2)) \geq 2R + \nu_2(m)$$

holds for nonzero even  $m$ , and also for odd  $m$  when  $R \geq 2$ . When  $m$  is odd and  $R = 1$ , replace the difference by a sum.

For the second family, the same parameter-refined bound holds whenever  $m$  is odd or  $R \geq 2$ . At the remaining first-descent case with  $m$  even, (7.1) gives the always-valid unsigned exponent  $\nu_2(m) + 1$ , while the signed statement retains exponent  $2 + \nu_2(m)$ . The ordinary exponent two holds without exception. The examples in Section 10 show why this distinction cannot simply be suppressed.

## 8 Framing as a change of coordinate

The preceding proof can be understood without geometric background. Nevertheless, its structure becomes more transparent when the normalized transforms are organized as a family of coordinate changes.

## 8.1 The additive law for the transformations

For an integral unit  $A$ , set

$$\mathfrak{T}_m(A)(y) = A(X_m(y)) =: A_m(y), \quad y = xA(x)^{-m}.$$

If a second transformation has parameter  $\ell$ , its coordinate is

$$z = yA_m(y)^{-\ell} = xA(x)^{-(m+\ell)}.$$

Uniqueness of formal inversion therefore proves

$$\mathfrak{T}_\ell(\mathfrak{T}_m(A)) = \mathfrak{T}_{m+\ell}(A), \quad \mathfrak{T}_0(A) = A. \quad (8.1)$$

Thus  $F = \mathfrak{T}_{-1}(A)$  is not an unrelated construction. It is one step in the same integer-parameter family. The shift from  $b_m$  to  $b_{m-1}$  in (4.8) reflects precisely this additive law.

## 8.2 The transformed potential

Define

$$\mathcal{W}_m(y) = W(X_m(y)) - \frac{m}{2}H(X_m(y))^2. \quad (8.2)$$

Using  $d \log y = (1 - mD(x))d \log x$ , differentiation of the right side gives

$$\delta_y \mathcal{W}_m(y) = H(X_m(y)), \quad \delta_y^2 \mathcal{W}_m(y) = \mathcal{B}_m(y).$$

All three series have zero constant term where appropriate, so

$$\mathcal{W}_m(y) = \sum_{n \geq 1} \frac{b_m(n)}{n^2} y^n. \quad (8.3)$$

The signed congruence theorem and Lemma 2.2 now imply an integral dilogarithm expansion

$$\mathcal{W}_m((-1)^m y) = \sum_{d \geq 1} c_{m,d} \operatorname{Li}_2(y^d), \quad (8.4)$$

where

$$c_{m,d} = \frac{1}{d^2} \sum_{e|d} \mu(d/e) (-1)^{me} b_m(e) \in \mathbb{Z}. \quad (8.5)$$

In particular,

$$A_m((-1)^m y) = \prod_{d \geq 1} (1 - y^d)^{-dc_{m,d}}. \quad (8.6)$$

The sign in this coordinate is invisible to odd-prime descent, but it is essential for the all-prime dilogarithm expansion.

This is the framing-integrality interpretation of the coefficient proof. Its general mathematical theory and geometric setting are the subject of [3, 4]. Higher-order congruences under more special framing hypotheses are studied in [5]; no such additional hypothesis is invoked here. In particular, the arithmetic of the input must not be assumed to transfer at every stronger order merely because second order transfers.

### 8.3 A second proof of the framing congruence: an exact Frobenius defect

The odd-prime part of Theorem 5.1 admits a second proof, logically independent of the constant-term computation of Section 5. Instead of working index by index in the original coordinate  $x$ , it produces a single exact series identity for the Frobenius defect of the framed potential  $\mathcal{W}_m$  in the framed coordinate  $y$ , from which integrality of  $b_m(n)$  and the congruence follow simultaneously. Both proofs are recorded because they have different strengths, compared at the end of this subsection. Nothing in Sections 5–7 depends on what follows.

Throughout, fix a prime  $p$  and an integer  $m$ , and write

$$x = X_m(y), \quad x_p = X_m(y^p), \quad A_m(y) = A(X_m(y)) \in 1 + y\mathbb{Z}[[y]]. \quad (8.7)$$

The series  $x_p$  and  $x^p$  are different, and the whole calculation turns on comparing them. Put

$$S(y) = \frac{1}{p}H(x_p) - H(x). \quad (8.8)$$

**Lemma 8.1** (Integral logarithmic error). *For every prime  $p$ , including 2, we have  $S(y) \in y\mathbb{Z}_{(p)}[[y]]$ , and*

$$x_p = x^p \exp(pmS(y)). \quad (8.9)$$

*Proof.* Since  $A_m$  has integer coefficients, reduction modulo  $p$  gives  $A_m(y^p) \equiv A_m(y)^p$ , so

$$\frac{A_m(y^p)}{A_m(y)^p} = 1 + p\Theta(y), \quad \Theta(y) \in y\mathbb{Z}_{(p)}[[y]].$$

Because  $H = \log A$ , we have  $\log A_m(y) = H(x)$  and  $\log A_m(y^p) = H(x_p)$ , whence

$$S(y) = \frac{1}{p} \log(1 + p\Theta(y)) = \sum_{j \geq 1} \frac{(-1)^{j+1} p^{j-1}}{j} \Theta(y)^j.$$

Each scalar  $p^{j-1}/j$  is  $p$ -integral: if  $e = \nu_p(j)$  then  $j \geq p^e \geq e + 1$ , so  $j - 1 - \nu_p(j) \geq 0$ . This includes  $p = 2$ . As  $\Theta$  has zero constant term, the sum is finite in each degree.

For (8.9), the coordinate equation (4.2) says  $x = yA_m(y)^m$ . Applying it at  $y^p$  and dividing by  $x^p = y^p A_m(y)^{pm}$  gives

$$\frac{x_p}{x^p} = \left( \frac{A_m(y^p)}{A_m(y)^p} \right)^m = \exp(pmS),$$

all quotients being quotients of formal units after cancelling the common leading monomial.  $\square$

The second ingredient is the logarithmic Taylor identity: for a series  $V$  in a variable  $\xi$  with zero constant coefficient, and a series  $q$  with zero constant term,

$$V(\xi e^q) = \sum_{k \geq 0} \frac{q^k}{k!} (\delta^k V)(\xi). \quad (8.10)$$

It suffices to check it on the monomials  $\xi^n$ , where both sides equal  $\xi^n e^{nq}$ ; one may then substitute series with zero constant term, and every coefficient involves only finitely many terms.

**Proposition 8.2** (Explicit defect formula). *Let*

$$\tilde{E}_{p,m}(y) = \mathcal{W}_m(y) - \frac{1}{p^2} \mathcal{W}_m(y^p). \quad (8.11)$$

Then, as an identity of formal series over  $\mathbb{Q}$ ,

$$\boxed{\begin{aligned} \tilde{E}_{p,m}(y) &= E_p(x) + mS \delta E_p(x) + \frac{m}{2} S^2 \\ &\quad - \sum_{k \geq 2} \frac{p^{k-2} m^k}{k!} S^k (\delta^k W)(x^p). \end{aligned}} \quad (8.12)$$

*Proof.* Apply (8.10) to  $V = W$  with  $\xi = x^p$  and  $q = pmS$ , which by (8.9) evaluates  $W$  at  $x_p$ . Dividing by  $p^2$  and using  $\delta W = H$  for the term  $k = 1$ ,

$$\frac{1}{p^2} W(x_p) = \frac{1}{p^2} W(x^p) + \frac{mS}{p} H(x^p) + \sum_{k \geq 2} \frac{p^{k-2} m^k}{k!} S^k (\delta^k W)(x^p).$$

By (8.2),

$$\tilde{E}_{p,m} = \left( W(x) - \frac{1}{p^2} W(x_p) \right) - \frac{m}{2} \left( H(x)^2 - (p^{-1} H(x_p))^2 \right).$$

The definition (8.8) gives  $p^{-1} H(x_p) = H(x) + S$ , so the second bracket equals  $-2H(x)S - S^2$  and its contribution is  $mH(x)S + \frac{m}{2} S^2$ . Substituting the displayed expansion into the first bracket turns it into

$$E_p(x) - \frac{mS}{p} H(x^p) - \sum_{k \geq 2} \frac{p^{k-2} m^k}{k!} S^k (\delta^k W)(x^p).$$

Adding the two contributions and using  $\delta E_p(x) = H(x) - p^{-1} H(x^p)$  from (5.3) proves (8.12).  $\square$

The series  $H(x)$  and  $p^{-1} H(x^p)$  need not be integral separately. Their cancellation into  $\delta E_p$  and  $S$  is exactly what prevents the denominator  $n$  in the logarithm from costing a power of the modulus. After it, only integral series and explicit factorial denominators remain.

**Theorem 8.3** (Framing congruence by the defect identity). *For every integer  $m$ ,*

$$\tilde{E}_{p,m}(y) \in \begin{cases} y\mathbb{Z}_{(p)}[[y]], & p \text{ odd}, \\ y\mathbb{Z}_{(2)}[[y]], & p = 2 \text{ and } m \text{ even}, \\ \frac{1}{2}y\mathbb{Z}_{(2)}[[y]], & p = 2 \text{ and } m \text{ odd}. \end{cases} \quad (8.13)$$

Consequently  $b_m(n) \in \mathbb{Z}$  for every integer  $m$  and every  $n \geq 1$ , and

$$b_m(N) \equiv b_m(N/p) \pmod{p^{2\nu_p(N)}}$$

whenever  $p \mid N$  and either  $p$  is odd or  $m$  is even; for  $p = 2$  and  $m$  odd this route yields  $\nu_2(b_m(N) - b_m(N/2)) \geq 2\nu_2(N) - 1$ .

*Proof.* By Lemma 2.3 and (5.3), the series  $E_p$  and  $\delta E_p$  lie in  $x\mathbb{Z}_{(p)}[[x]]$ ; by Lemma 8.1,  $S \in y\mathbb{Z}_{(p)}[[y]]$ ; by (2.8),  $\delta^k W \in x\mathbb{Z}[[x]]$  for  $k \geq 2$ . Composition preserves integrality because  $x \in y\mathbb{Z}[[y]]$  and  $x^p \in y\mathbb{Z}[[y]]$ .

For odd  $p$  the scalar  $1/2$  is  $p$ -integral, and (5.9) makes every scalar  $p^{k-2}m^k/k!$   $p$ -integral. Hence the right side of (8.12) lies in  $y\mathbb{Z}_{(p)}[[y]]$ . For  $p = 2$  the same estimates allow at worst one factor  $1/2$ , arising both from  $\frac{m}{2}S^2$  and from the factorial scalars. If  $m$  is even and nonzero, then for  $k \geq 2$

$$\nu_2 \left( \frac{2^{k-2} m^k}{k!} \right) \geq (k-2) + k - (k-1) = k-1 \geq 1,$$

and  $\frac{m}{2}S^2$  is integral as well; if  $m = 0$ , all these terms vanish. This proves (8.13).

Now extract coefficients, *without* assuming in advance that  $b_m(n)$  is integral. By (8.3),

$$[y^N]\tilde{E}_{p,m}(y) = \begin{cases} b_m(N)/N^2, & p \nmid N, \\ (b_m(N) - b_m(N/p))/N^2, & p \mid N. \end{cases} \quad (8.14)$$

When the defect is integral, the first case shows that  $b_m(N)$  is  $p$ -integral for  $p \nmid N$ , and the second case then proves by induction on  $\nu_p(N)$  that every  $b_m(N)$  is  $p$ -integral, while simultaneously giving  $\nu_p(b_m(N) - b_m(N/p)) \geq 2\nu_p(N)$ . This covers every odd prime and also  $p = 2$  with  $m$  even.

For  $p = 2$  and  $m$  odd, the defect is only half-integral, and (8.14) yields exactly the weaker bound  $2\nu_2(N) - 1$ . Integrality at 2 is nonetheless available in this case from the quotient  $b_m(n) = \frac{1}{m}[x^n]A(x)^{mn}$  of Proposition 4.1, whose numerator is an integer, because  $A^{mn}$  is an integer power of an integral unit, and whose denominator  $m$  is odd. The case  $m = 0$  is Proposition 3.1. Thus every  $b_m(n)$  is integral at every prime, hence lies in  $\mathbb{Z}$ .  $\square$

**Remark 8.4** (The stated dyadic bound is the sharper one). Theorem 8.3 is a correct lower bound, not an error, but for  $p = 2$  with odd  $m$  it is superseded. The constant-term argument of Section 6 improves  $2\nu_2(N) - 1$  to the full precision  $2\nu_2(N)$  as soon as  $\nu_2(N) \geq 2$ , see (6.4), and at the first dyadic descent it replaces the difference by the exact signed sum (6.5), which the defect route does not see at all. Everywhere the two disagree, the value asserted by this article is the one proved in Section 6. Likewise, for the second family with nonzero even  $m$  and  $\nu_2(N) \geq 2$ , the bound recorded in this article is  $2\nu_2(N) + \nu_2(m)$ , from Section 7, rather than the  $2\nu_2(N) + \nu_2(m) - 1$  that the defect route alone gives. At  $\nu_2(N) = 1$  the two agree: both give the exponent  $\nu_2(m) + 1$ , and (7.1) together with the example  $\nu_2(2) - \nu_2(1) = 236$  shows that this is attained.

**Remark 8.5** (What each route buys). The constant-term route of Sections 5–6 works index by index in the original coordinate  $x$ ; its engine is the integration by parts (5.8), which recovers the second power of  $N$  in the linear term. Because it keeps the exact Laurent identity (6.9) in reach, it is what makes the delicate dyadic sign analysis, and hence the signed all-prime Theorem 5.1, accessible. The defect route of this subsection replaces that bookkeeping by one exact series identity (8.12) in the framed coordinate; it yields integrality of  $b_m(n)$  and the congruence together by induction on  $\nu_p(N)$  at every odd prime, and at  $p = 2$  for even  $m$ , without appealing to the separate integral definition (4.3); only the residual  $p = 2$ ,  $m$  odd case is still read off from the quotient of Proposition 4.1, it can be checked symbolically as a formal identity rather than only through its numerical consequences, and it is the form that transparently transfers to other admissible seeds, as in Section 9. Neither route is a specialization of the other.

## 9 An infinite family of admissible seeds

Both proofs of the framing congruence used only that the input sequence is an integer sequence satisfying Definition 2.1. The Apéry numbers are not an isolated such input: the elementary argument of Proposition 3.1 extends verbatim to every even binomial moment.

**Proposition 9.1** (Binomial moments). *For every positive integer  $h$ , the sequence*

$$a_n^{(h)} = \sum_{k=0}^n \left( \binom{n}{k} \binom{n+k}{k} \right)^{2h} = \sum_{k=0}^n P(n, k)^{2h}$$

*satisfies the second-order Gauss congruences of Definition 2.1. Consequently Theorem 5.1 and Theorem 1.1 hold verbatim with  $a_n^{(h)}$  in place of  $a_n$ , for the exponential family and for its reversion alike.*

*Proof.* Fix a prime  $p$ , a positive  $N$  divisible by  $p$ , and put  $R = \nu_p(N)$ . If  $p \nmid k$ , then  $\nu_p(P(N, k)) \geq R$  as in Proposition 3.1, so  $\nu_p(P(N, k)^{2h}) \geq 2hR \geq 2R$  and the summand vanishes modulo  $p^{2R}$ . If  $k = p\ell$ , raise the split product (3.3) to the even power  $2h$ : the sign disappears and each remaining factor lies in  $1 + p^{2R}\mathbb{Z}_{(p)}$ , so  $P(N, p\ell)^{2h} \equiv P(N/p, \ell)^{2h} \pmod{p^{2R}}$ . The term  $k = 0$  is one on both sides. Summing over  $k$  gives  $a_N^{(h)} \equiv a_{N/p}^{(h)} \pmod{p^{2R}}$ . With this input, the proofs of Sections 4–7 and of Section 8.3 apply without change.  $\square$

The case  $h = 1$  is the classical Apéry sequence used to resolve the two selected conjectures. The extension is included to show what the argument actually depends on: an arithmetic property of a binomial sum and an integral change of variable, not a numerical fit or an unproved classification of recurrences.

## 10 Examples and limits of stronger assertions

### 10.1 Low-degree values and exceptional parameters

Direct expansion gives

$$\begin{aligned} A(x) &= 1 + 5x + 49x^2 + 685x^3 + 11807x^4 + 232771x^5 + O(x^6), \\ F(x) &= 1 + 5x + 24x^2 + 200x^3 + 2430x^4 + 36096x^5 + O(x^6). \end{aligned}$$

These values agree with the target entry [1]. The normalized coefficients are already nontrivial polynomials in the parameter:

$$\begin{aligned} b_m(1) &= 5, \\ b_m(2) &= 73 + 50m, \\ b_m(3) &= \frac{2890 + 3285m + 1125m^2}{2}, \\ b_m(4) &= \frac{99003 + 147574m + 87600m^2 + 20000m^3}{3}. \end{aligned} \tag{10.1}$$

The last two expressions have rational polynomial coefficients but take integer values at every integer  $m$ , as guaranteed structurally by (4.3). One should not confuse an integer-valued polynomial with a polynomial all of whose coefficients are integers.

Expanding the exponential gives the finite closed expression

$$b_m(n) = \sum_{\substack{k_1, \dots, k_n \geq 0 \\ k_1 + 2k_2 + \dots + nk_n = n}} n^K m^{K-1} \prod_{j=1}^n \frac{(a_j/j)^{k_j}}{k_j!}, \quad K = k_1 + \dots + k_n. \tag{10.2}$$

Since  $n \geq 1$ , every summand has  $K \geq 1$ ; at  $m = 0$  only  $K = 1$  survives and that term is  $a_n$ , in agreement with  $b_0(n) = a_n$ . Formula (10.2) is convenient for symbolic experimentation, but its denominators do not by themselves explain either integrality or the supercongruences.

| $m$ | $b_m(1)$ | $b_m(2)$ | $b_m(3)$ | $b_m(4)$ | $b_m(5)$ |
|-----|----------|----------|----------|----------|----------|
| −2  | 5        | −27      | 410      | −1915    | 53130    |
| −1  | 5        | 23       | 365      | 6343     | 123255   |
| 0   | 5        | 73       | 1445     | 33001    | 819005   |
| 1   | 5        | 123      | 3650     | 118059   | 4015380  |
| 2   | 5        | 173      | 6980     | 301517   | 13540505 |
| 3   | 5        | 223      | 11435    | 623375   | 35175630 |

The cases  $m = 0$  and  $m = 1$  now have concrete interpretations. The normalized family at  $m = 0$  is the Apéry sequence, but the original  $u_0$  and  $v_0$  vanish at positive indices. The reversion identity  $v_1(n) = a_n$  is an exact coefficient identity, not just a congruence. Furthermore,  $v_2(n) = 2u_1(n)$  for every positive  $n$ .

For comparison, some primitive coefficients from (8.5) are

| $m$ | $c_{m,1}$ | $c_{m,2}$ | $c_{m,3}$ | $c_{m,4}$ |
|-----|-----------|-----------|-----------|-----------|
| -1  | -5        | 7         | -40       | 395       |
| 0   | 5         | 17        | 160       | 2058      |
| 1   | -5        | 32        | -405      | 7371      |
| 2   | 5         | 42        | 775       | 18834     |

Negative coefficients present no difficulty: integrality, rather than positivity, is the property under consideration.

## 10.2 The first unsigned family cannot include two unconditionally

At  $m = 1$ ,  $N = 2$ ,

$$u_1(2) - u_1(1) = 123 - 5 = 118 \not\equiv 0 \pmod{4}.$$

Thus a blanket unsigned extension of the first OEIS assertion to  $p = 2$  is false. The signed statement instead gives

$$u_1(2) + u_1(1) = 128 \equiv 0 \pmod{4},$$

exactly as predicted by the first-descent proof.

The stronger parameter factor at two also has a genuine boundary for the second family:

$$v_2(2) - v_2(1) = 246 - 10 = 236 \equiv 0 \pmod{4}, \quad 236 \not\equiv 0 \pmod{8}.$$

Since  $2R + \nu_2(m) = 3$  in this example, the unsigned parameter-refined bound cannot hold uniformly at two. The original second-order congruence still holds, and the signed sum is 256, divisible by 8.

## 10.3 Second order cannot uniformly be replaced by third order

A small odd-prime example also limits the possible strengthening. Exact coefficient extraction gives  $u_1(7) = 5\,082\,313\,276$ , hence

$$\begin{aligned} u_1(7) - u_1(1) &= 5082313271 \\ &= 7^2 \cdot 103720679, \quad 103720679 \equiv 6 \pmod{7}, \end{aligned} \tag{10.3}$$

so that  $u_1(7) - u_1(1) \equiv 294 \pmod{343}$ . By  $v_2(n) = 2u_1(n)$ ,

$$v_2(7) - v_2(1) = 10\,164\,626\,542 \equiv 245 \pmod{343},$$

again of valuation exactly two. The valuation is exactly two, not three. Therefore a universal replacement of  $p^{2r}$  by  $p^{3r}$  in the first conjecture is false. Because  $v_2(n) = 2u_1(n)$  and  $7 \nmid 2$ , the same example rules out that replacement for the second family as well. This is a counterexample to a uniform higher-order assertion, not a claim that no individual parameters or primes can exhibit stronger divisibility. These are counterexamples to stronger variants, not to either formula actually listed in OEIS.

## 11 Exact computation and reproducibility

Two independent companion programs accompany this article, one written for each of the two proofs. Both use only the Python standard library and exact arithmetic, and every division that is mathematically supposed to be exact is checked with a remainder test.

| Program                                    | What it tests                                                                                                                                                                                                        |
|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>verify_constant_term_route.py</code> | Arbitrary-precision integers. Index range to 250, signed and unsigned normalized congruences, direct degree-12 polynomial powers, both directions of compositional inversion. Output in <code>verification/</code> . |
| <code>verify_defect_route.py</code>        | Integers and <code>Fraction</code> . Framing parameters $-10$ to $10$ , and 18 symbolic checks of the defect identity (8.12) itself through degree 14. Output in <code>verification-defect-route/</code> .           |

They are deliberately not refactored into one program: they compute the same quantities by different routes, and their independence is the point. The general proofs above are independent of both; the programs check identities, edge cases, and representative congruences by routes that expose indexing or normalization mistakes.

### 11.1 Coefficient algorithms

For a fixed integer  $q$ , write  $A(x)^q = \sum_{j \geq 0} h_j x^j$ . Logarithmic differentiation gives

$$h_0 = 1, \quad h_j = \frac{q}{j} \sum_{k=1}^j a_k h_{j-k} \quad (j \geq 1). \quad (11.1)$$

The quotient is an integer because  $A^q$  is an integral unit. This recurrence is valid for positive, zero, and negative  $q$ . To obtain  $u_m(N)$ , use the *fixed* exponent  $q = mN$  throughout the computation through degree  $N$ . The exponent must not be changed to  $mj$  as the internal coefficient index  $j$  varies.

For  $m \neq 0$ , compute  $b_m(N)$  by exact division of  $h_N$  by  $m$ ; for  $m = 0$ , use  $a_N$ . Then compute  $v_m(N) = mb_{m-1}(N)$ . This requires  $O(N^2)$  integer arithmetic operations and  $O(N)$  stored integers for one coefficient at a fixed parameter and target index. These are arithmetic-operation bounds, not unit-cost bit-complexity claims for the large integers involved.

The input numbers are generated directly from the binomial definition using

$$P(n, 0) = 1, \quad P(n, k) = P(n, k-1) \frac{(n-k+1)(n+k)}{k^2}, \quad a_n = \sum_{k=0}^n P(n, k)^2.$$

They are independently compared against the recurrence recorded in A005259 [2]:

$$(n+1)^3 a_{n+1} = (34n^3 + 51n^2 + 27n + 5)a_n - n^3 a_{n-1}, \quad a_0 = 1, \quad a_1 = 5. \quad (11.2)$$

The leading polynomial is often written in the equivalent factored form  $(2n+1)(17n^2+17n+5)$ ; the second program uses that spelling. This familiar recurrence is used only as a cross-check; neither its validity nor a proof of it is needed for the congruence argument.

For independent low-degree reversion checks, Lagrange inversion gives

$$[y^n]F(y) = \begin{cases} 5, & n = 1, \\ \frac{1}{1-n} [x^n]A(x)^{1-n}, & n \geq 2. \end{cases}$$

The program verifies  $F(y) = A(y/F(y))$ , both directions of compositional inversion, and direct polynomial powers of  $A$  and  $F$  through degree 12. These direct power checks do not assume the formulas  $u_m = mb_m$  and  $v_m = mb_{m-1}$  as their coefficient-computation method.

## 11.2 The first included run: the constant-term program

The archived run of `verify_constant_term_route.py` used maximum index 250, original parameters  $-6 \leq m \leq 6$ , and normalized parameters  $-7 \leq m \leq 6$ . For the transformed congruences it used primes 2, 3, 5, 7, 11, 13 and indices  $N = hp^r \leq 250$  with  $1 \leq h \leq 6$  and  $r \geq 1$ . Duplicate prime-index pairs were removed, leaving 71 pairs. This is a specified finite sample, not every possible transformed parameter and index up to 250.

The input Apéry congruences were checked more exhaustively: every prime divisor of every positive  $N \leq 250$  was used. The completed run reported **8,142 successful checks**. The grouped counts are:

| Check category                                             | Count |
|------------------------------------------------------------|-------|
| Apéry definition versus three-term recurrence              | 251   |
| Input Apéry second-order congruences                       | 473   |
| Input primitive dilogarithm coefficients                   | 250   |
| Functional equation, two inverse identities, Euler product | 4     |
| Direct coefficient identities for $u_m$ and $v_m$          | 312   |
| Signed normalized congruences at all selected primes       | 994   |
| Unsigned normalized congruences in the applicable cases    | 973   |
| Unsigned $v_m$ congruences, including the prime two        | 923   |
| Unsigned $u_m$ congruences for odd primes or even $m$      | 815   |
| Signed parameter-refined congruences for both families     | 1704  |
| Odd-prime parameter-refined congruences for both families  | 1272  |
| Signed transformed primitive dilogarithm coefficients      | 168   |
| Boundary and higher-order counterexamples                  | 3     |
| Total                                                      | 8142  |

The JSON report gives all 71 prime-index pairs, the ungrouped category counts, the exact counterexample values, and the arithmetic environment. The CSV file records sample values of the sequences. No floating-point approximation or probabilistic primality test is used.

## 11.3 The second included run: the defect-identity program

The program `verify_defect_route.py` was written for the proof of Section 8.3 and uses a different grid. It generates the seed by the recurrence (11.2) and compares it against an independent evaluation of the defining binomial sum at every index. It constructs the compositional inverse of  $xA(x)$  by solving its coefficients successively, checks the composition, forms  $F = y/X_{-1}(y)$ , and computes the powers  $F^{mn}$  by rational polynomial convolution, so that those tests do not generate the left side of  $v_m(n) = mb_{m-1}(n)$  from the identity being tested. Finally it constructs both sides of (8.12) as truncated rational series through degree 14, which tests the sign of the quadratic correction, the factor  $p^{k-2}$ , and the placement of  $x^p$  as distinct from  $x_p$ , rather than only comparing a selection of final congruences.

| Check category                                          | Count |
|---------------------------------------------------------|-------|
| Binomial sum versus recurrence, indices 0–150           | 151   |
| Seed congruences, all primes and multiples $N \leq 150$ | 269   |
| Odd-prime normalized congruences on the specified grid  | 798   |
| Normalized dyadic bounds, all even $N \leq 150$         | 1575  |
| Parameter-sensitive odd-prime bounds for $u_m$          | 840   |
| Parameter-sensitive dyadic bounds for $u_m$             | 1575  |
| Bounds for $v_m$ , including the prime two              | 2415  |
| Independent reversion-and-power identities              | 126   |
| Full Frobenius-defect identities through degree 14      | 18    |
| Total                                                   | 7767  |

The normalized odd-prime grid uses  $m = -10, -9, \dots, 10$ ,  $p \in \{3, 5, 7, 11, 13\}$ ,  $1 \leq r \leq 4$ ,  $1 \leq n \leq 4$ , retaining  $N = np^r \leq 150$ . The dyadic normalized tests use all even  $N \leq 150$  for the same parameter range. The independent reversion tests use  $m = -4, \dots, 4$  and  $1 \leq n \leq 14$ , and the 18 formal defect tests use  $m \in \{-3, -1, 0, 1, 2, 4\}$  and  $p \in \{2, 3, 5\}$ . The dyadic bounds tested by this program are the ones its own proof yields, namely  $2\nu_2(N) - 1$  for odd framing parameters; the sharper bounds asserted in Theorem 1.1 and Theorem 5.1 are tested by the first program.

The two runs are different finite samples, not comparable totals: the first reaches higher indices, the second reaches wider parameters and is the only one that checks its central identity symbolically. Neither is an exhaustive search.

## 11.4 Reproduction commands

From the archive directory, with Python 3.9 or later, run

```
python verify_constant_term_route.py --max-index 250 \
--output-dir verification
```

A larger finite test can be requested, for example, with

```
python verify_constant_term_route.py --max-index 500 \
--parameter-radius 8 --multipliers 8 \
--primes 2 3 5 7 11 13 17 --output-dir verification-500
```

The second program requires Python 3.10 or later and is run as

```
python verify_defect_route.py --limit 150 \
--out verification-defect-route
```

Run it normally, not with the `-O` option: it uses assertions. Only the default runs described above are represented by the archived verification results. To rebuild the article, use

```
latexmk -pdf -interaction=nonstopmode -halt-on-error article.tex
```

The source is also compatible with two ordinary `pdflatex` runs. The computational artifact is not a proof-assistant formalization; the proof is the mathematical argument in this article. Numerical checks cannot prove an infinite family of congruences. Their role here is to make the examples reproducible and to expose implementation, normalization, or sign errors in independently computed quantities.

## 12 Conclusion

The two parameterized assertions in A267220 follow from the same normalized transform, with a shift of one in the framing parameter. The central arithmetic fact is preservation of second-order Gauss congruences after this normalization. Its coefficient proof isolates the only linear term that appears to lose one power of the index and recovers that power by formal integration by parts. The same preservation is proved a second time, independently, as the single exact identity (8.12) for the Frobenius defect in the framed coordinate; that route yields integrality and the congruence at once and transfers unchanged to the infinite family of binomial-moment seeds of Proposition 9.1, while the constant-term route is what reaches the dyadic sign correction.

For the stated OEIS conjectures the conclusion is unconditional: every integer parameter, every allowed prime, and every positive pair of indices is covered. In fact, all odd primes work with an additional parameter valuation, and the second family works unsigned at two. The dyadic sign correction is exact, not an artifact of an insufficient estimate, as the smallest counterexample demonstrates. Finally, second order is genuinely the strongest order that can be asserted uniformly for these families without additional restrictions.

## A A formal Lagrange inversion proof

For completeness, the coefficient formula used above is established here in the precise formal setting required.

**Lemma A.1.** *Let  $\phi(x) \in 1 + x\mathbb{Q}[[x]]$  and let  $X(y)$  be the unique series satisfying  $X = y\phi(X)$ . For any  $G(x) \in \mathbb{Q}[[x]]$  and  $n \geq 1$ ,*

$$[y^n]G(X(y)) = \frac{1}{n}[x^{n-1}]G'(x)\phi(x)^n. \quad (\text{A.1})$$

*Proof.* Let  $\text{Res}_x$  select the coefficient of  $x^{-1}$  from a formal differential. Residues are unchanged by a substitution  $y = x + O(x^2)$ . This can be checked on monomials: derivatives have zero residue, and  $y'/y = x^{-1}$  plus a power series.

Substitute  $y = x/\phi(x)$  in the coefficient residue. Then

$$\begin{aligned} [y^n]G(X(y)) &= \text{Res}_y G(X(y))y^{-n-1} dy \\ &= \text{Res}_x G(x)x^{-n-1}\phi(x)^n \left(1 - \frac{x\phi'(x)}{\phi(x)}\right) dx. \end{aligned}$$

The residue of  $d(G(x)x^{-n}\phi(x)^n)$  is zero. Expanding that derivative and rearranging shows that the last display equals

$$\frac{1}{n} \text{Res}_x G'(x)x^{-n}\phi(x)^n dx = \frac{1}{n}[x^{n-1}]G'(x)\phi(x)^n,$$

as required. □

## B Proof dependencies and boundary cases

For an independent audit, the logical dependencies can be summarized without treating computation as a lemma. The binomial identity (3.2) proves Proposition 3.1. Möbius inversion proves Lemma 2.2, including integrality of  $A$ . Integral compositional inversion and the Lagrange formula identify the

coefficients of  $\mathcal{B}_m$  and  $\mathcal{W}_m$ . The constant-term identity (5.8), the factorial bounds, and the three dyadic cases prove Theorem 5.1; alternatively, the logarithmic lemma, the exact Taylor cancellation, and the same factorial bounds prove its odd-prime part through Theorem 8.3. The parameter-shift identity (4.8) then proves both OEIS statements.

The points most vulnerable to an incorrect shortcut are the following. The inverse is *compositional*, and  $F = A \circ X_{-1}$ , not  $A \circ X_1$ ; it is not a multiplicative reciprocal. The quadratic correction in (8.2) has sign  $-m/2$  for the coordinate  $y = xA(x)^{-m}$ . The Frobenius Taylor expansion is taken at  $x^p$ , whereas  $x_p$  denotes  $X_m(y^p)$ ; these are generally different, and (8.9) is exactly their comparison. The normalized coefficient  $b_m(n)$  is not assumed integral before it is proved so. Division by  $m - 1$  is avoided at  $m = 1$  and is never performed modulo  $p$ , and division by  $m$  is never performed modulo  $p$  either. Negative parameters are integer powers of units and need no positivity assumption. The dyadic half-integrality of the framed defect is retained rather than silently replaced by integrality, and where the two routes give different dyadic exponents the sharper proved one is the one asserted, as recorded in Remark 8.4. Finally, the modulus depends on the full valuation of  $N$ , not on a representation of  $N$  with a tacitly prime-to- $p$  leading factor.

At index zero the original definitions give  $u_m(0) = v_m(0) = 1$ , even for  $m = 0$ . All normalized identities in this article are stated for positive indices. The conjectures themselves have positive  $n$  and  $r$ , so this convention causes no exceptional case in their application.

## References

- [1] OEIS Foundation Inc., *A267220: Expansion of  $\exp(\sum_{n \geq 1} A005259(n)x^n/n)$* . The two supercongruence conjectures are attributed to Peter Bala, October 17, 2024. Entry checked September 20, 2026. <https://oeis.org/A267220>.
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- [4] A. Schwarz, V. Vologodsky, and J. Walcher, *Integrality of Framing and Geometric Origin of 2-functions (with algebraic coefficients)*, arXiv:1702.07135, version 2 (2017), especially Section 3. <https://arxiv.org/abs/1702.07135>.
- [5] L. Felipe Müller, *Wolstenholme Type Congruences and Framing of Rational 2-Functions*, arXiv:2104.10754 (2021). <https://arxiv.org/abs/2104.10754>.
- [6] J.-C. Liu, *An extension of Gauss congruences for Apéry numbers*, arXiv:2404.16636v1 (25 April 2024). <https://arxiv.org/abs/2404.16636>. Used for background, the standard recurrence, and the historical order-three congruences of Gessel and Coster; the input order-two congruence is proved directly in this article, so this reference is context and not a dependency.